

Chapter 9.A

Periodicity and Ionic Bonding

Chemical Principles I

CHEM 1211K

1. Relate electron configuration and position in the periodic table to the number of valence electrons of an element.
2. Apply Slater's rules to determine effective nuclear charge (Z_{eff}).
3. Discuss the periodic trends of atomic and ionic radius based on the concept of effective nuclear charge (Z_{eff}).
4. Predict the relative ionization energies of different elements from their positions in the periodic table.
5. Predict the relative electron affinities of different elements from their positions in the periodic table.
6. Determine the charges of common monatomic ions based on the positions of their elements on the periodic table.
7. Describe how electrostatic attractive forces allow ionic compounds to exist as very large, three-dimensional lattices.
8. Describe and apply periodic trends in lattice energy.

9.1 Core and Valence Electrons



- **Core electrons** are in the inner shells of the atom; core electrons remain close to the nucleus and do not participate in bonding or chemical reactions.
- **Valence electrons** are in the outermost shell of the atom.

9.1 Counting Valence Electrons

- For the main-group elements, the number of valence electrons in the neutral atom can be inferred from the group number.

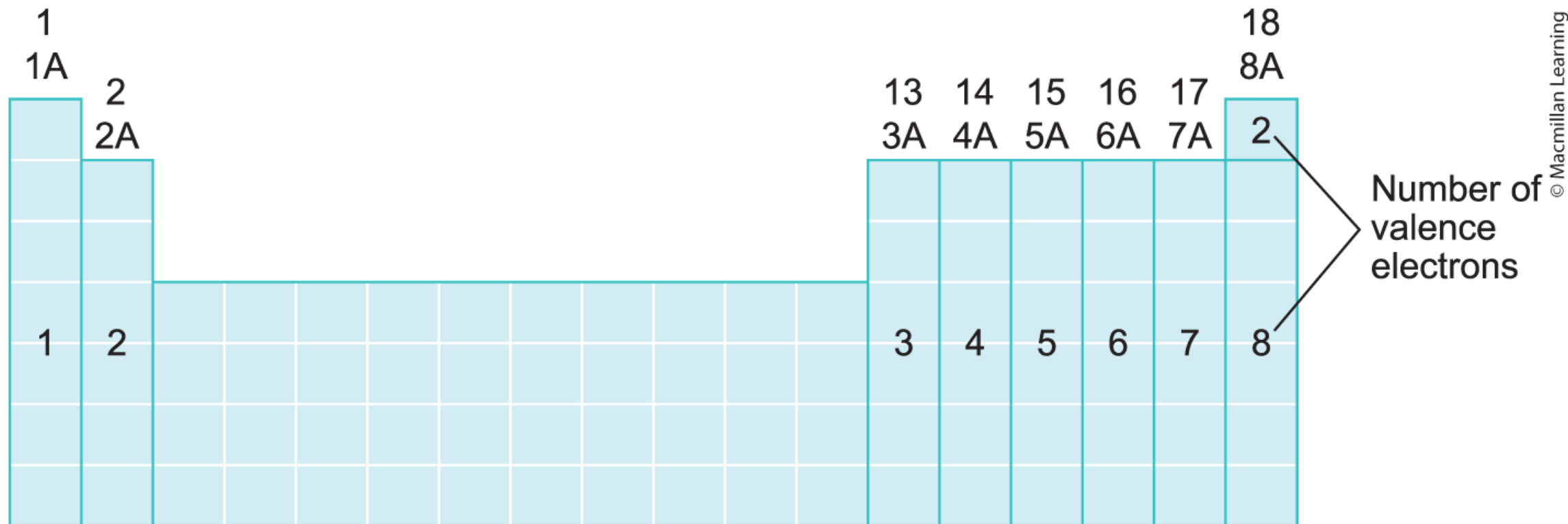
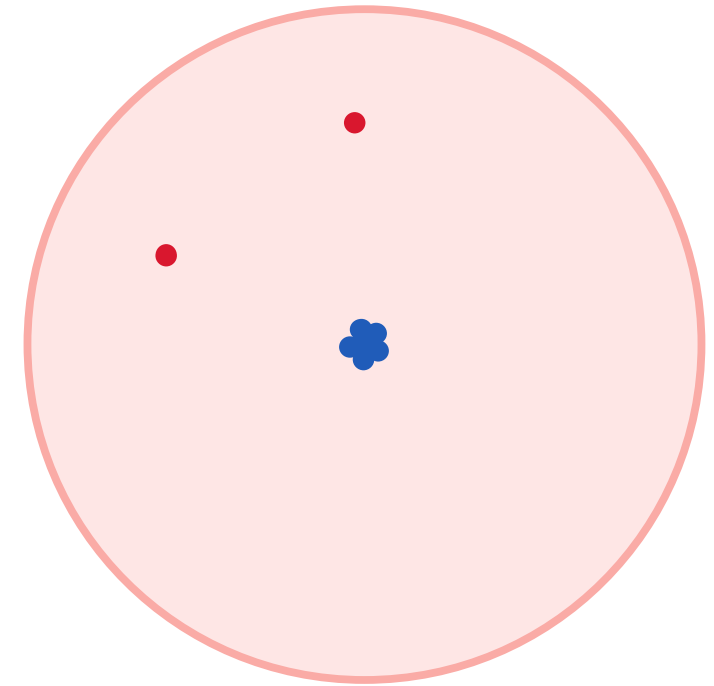


Figure 9.1. Valence electron counts for the main-group elements.

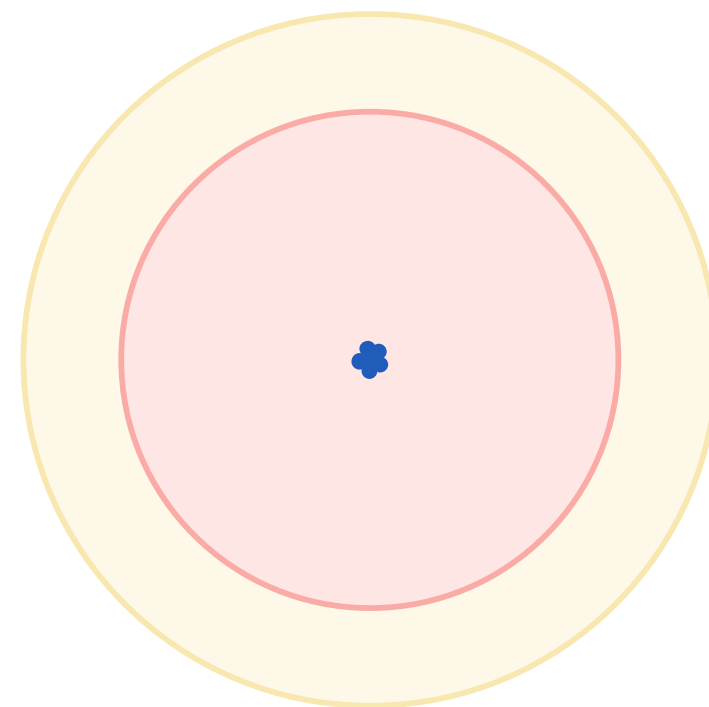
9.2 Attraction and Repulsion in Atoms

- The sizes of atoms and ions are dictated by two types of fundamental forces in the atom:
 - 1.
 - 2.
- Interactions between the electrons and nucleus follow principles of **electrostatics...**



9.2 Effective Nuclear Charge (Z_{eff})

- The valence electrons do not “feel” the full attractive force of the nucleus due to repulsion from other valence electrons and (more important) core electrons.
- **Effective nuclear charge (Z_{eff})** is the net positive charge felt by an electron after accounting for electron-electron repulsion.
- In this context, electron-electron repulsion that reduces the value of Z_{eff} is called **shielding**.



9.2 Slater's Rules for Calculating Shielding

- **Slater's rules** are a set of conventions for determining the value of the shielding constant S for the last valence electron. Only the numbers of core (N_{core}) and valence electrons (N_{val}) are used; distinctions are not made between different core shells.

$$S =$$

$$Z_{eff} =$$

9.2 Determining Effective Nuclear Charge



Example. Calculate the effective nuclear charge felt by the outermost valence electron in a neutral neon atom.

9.2 Atomic Radius

- **Atomic radius** increases down a group and generally decreases from left to right across a period.
- Note that atomic radius is *negatively* correlated with Z_{eff} across a period.

Element	Z_{eff}	Atomic radius (pm)
Na	2.50	160
Mg	3.15	140
Al	3.80	124
Si	4.45	114
P	5.10	109
S	5.75	104
Cl	6.40	100

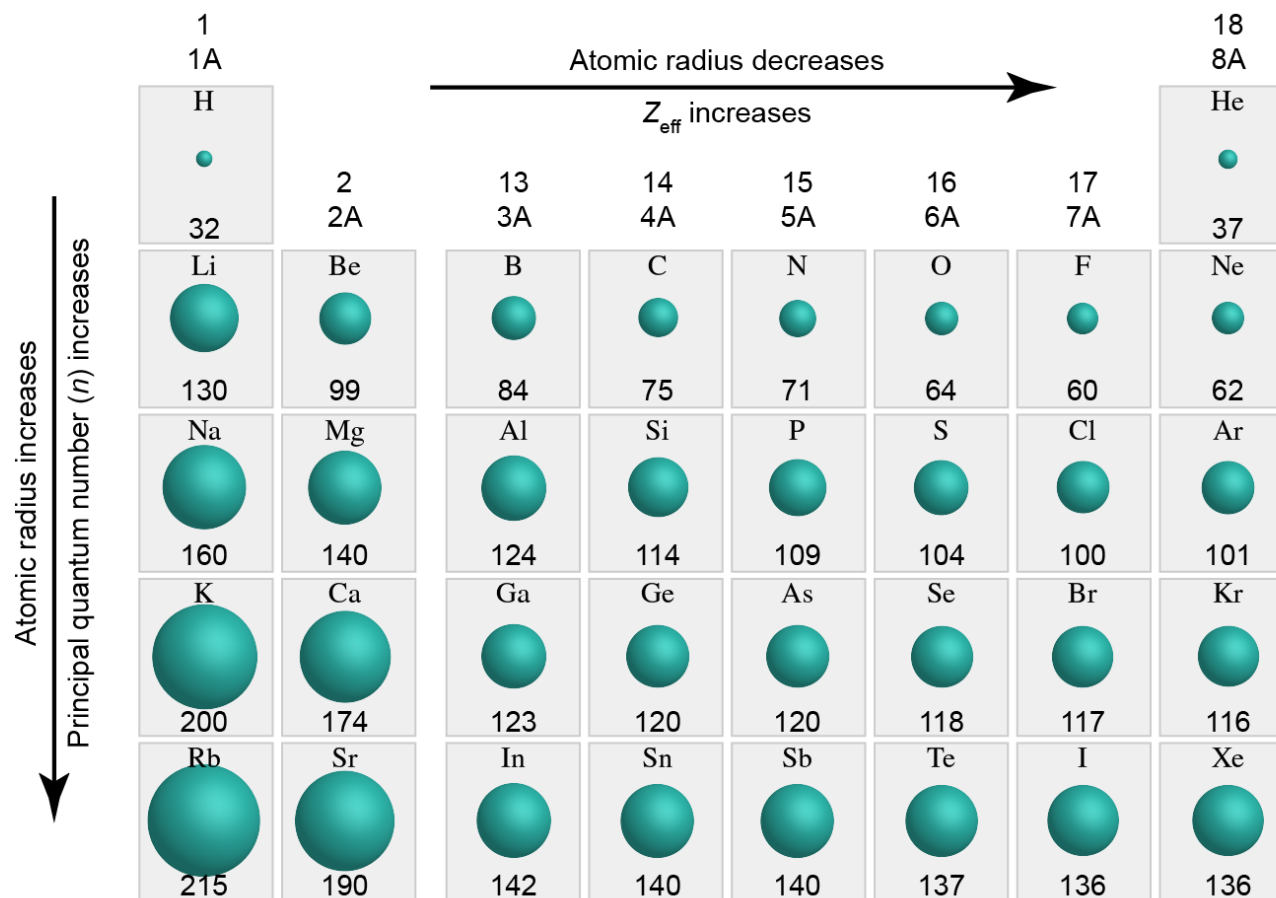


Figure 9.3. Atomic radii for main-group elements.

Atoms A and B are in the same period and atom A has a smaller atomic radius than atom B. Which statement *best* explains why this is true?

- A. The valence electrons of atom B are more strongly attracted to its nucleus than are the valence electrons of atom A to its nucleus.
- B. Atom A has fewer protons than atom B.
- C. Z_{eff} for A (for the outermost valence electron) is greater than Z_{eff} for B.
- D. Atom A has more core electrons than B.
- E. Atom B has more valence electrons than A.

Arrange these atoms in order of increasing radius:

- A. Nitrogen
- B. Oxygen
- C. Sodium
- D. Phosphorus

Enter your response as a string of letters from smallest to largest, such as “ABCD.”

9.2 Cations and Anions Relative to Neutral Atoms

- All cations have smaller radii than their corresponding neutral atoms.
- All anions have larger radii than their corresponding neutral atoms.

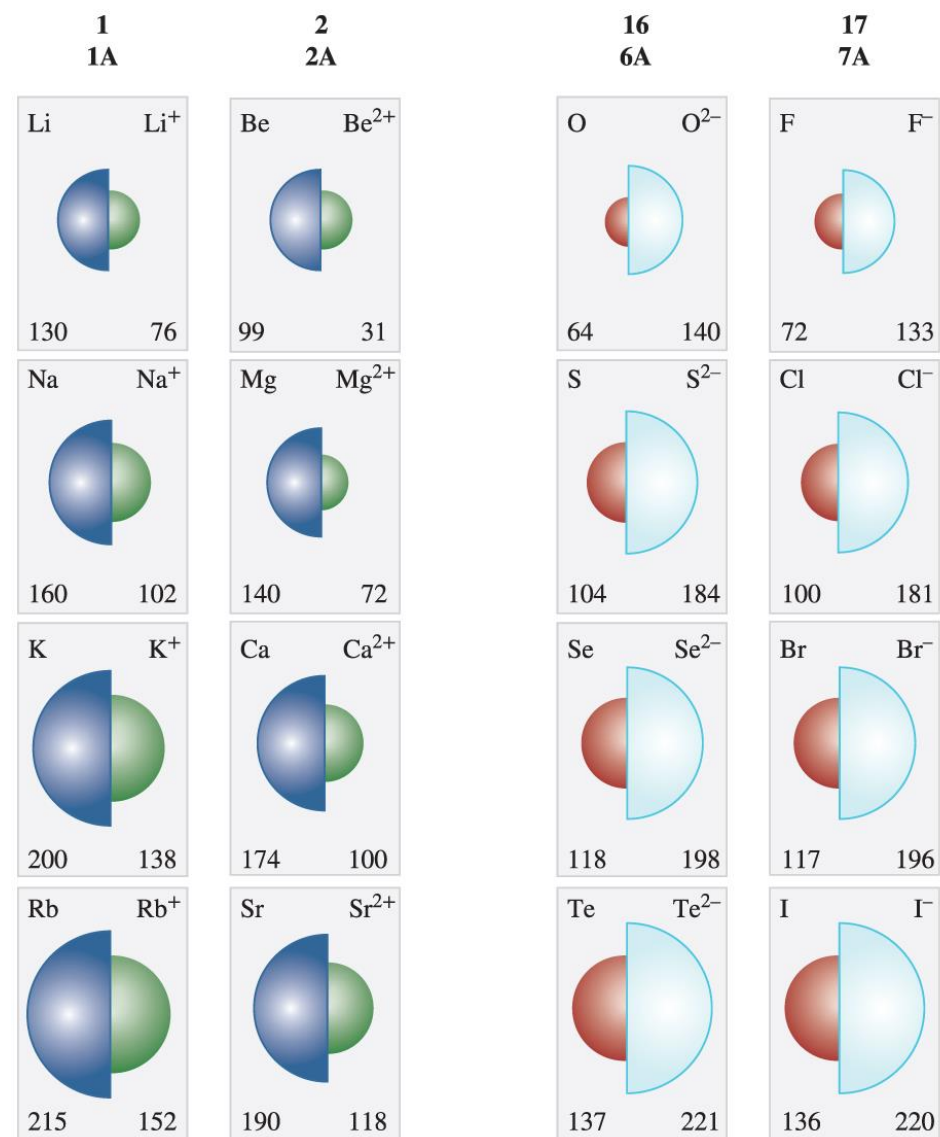


Figure 9.5. Radii of neutral atoms and stable monatomic ions.

- When electrons are lost or gained from a neutral atom to form an ion, the numbers of protons and core electrons remain constant, but the number of valence electrons changes. Z_{eff} changes as a result.
- Cations and anions decrease in radius left to right across a period and increase in radius down a group, just like neutral atoms.

9.2 Ionic Radii

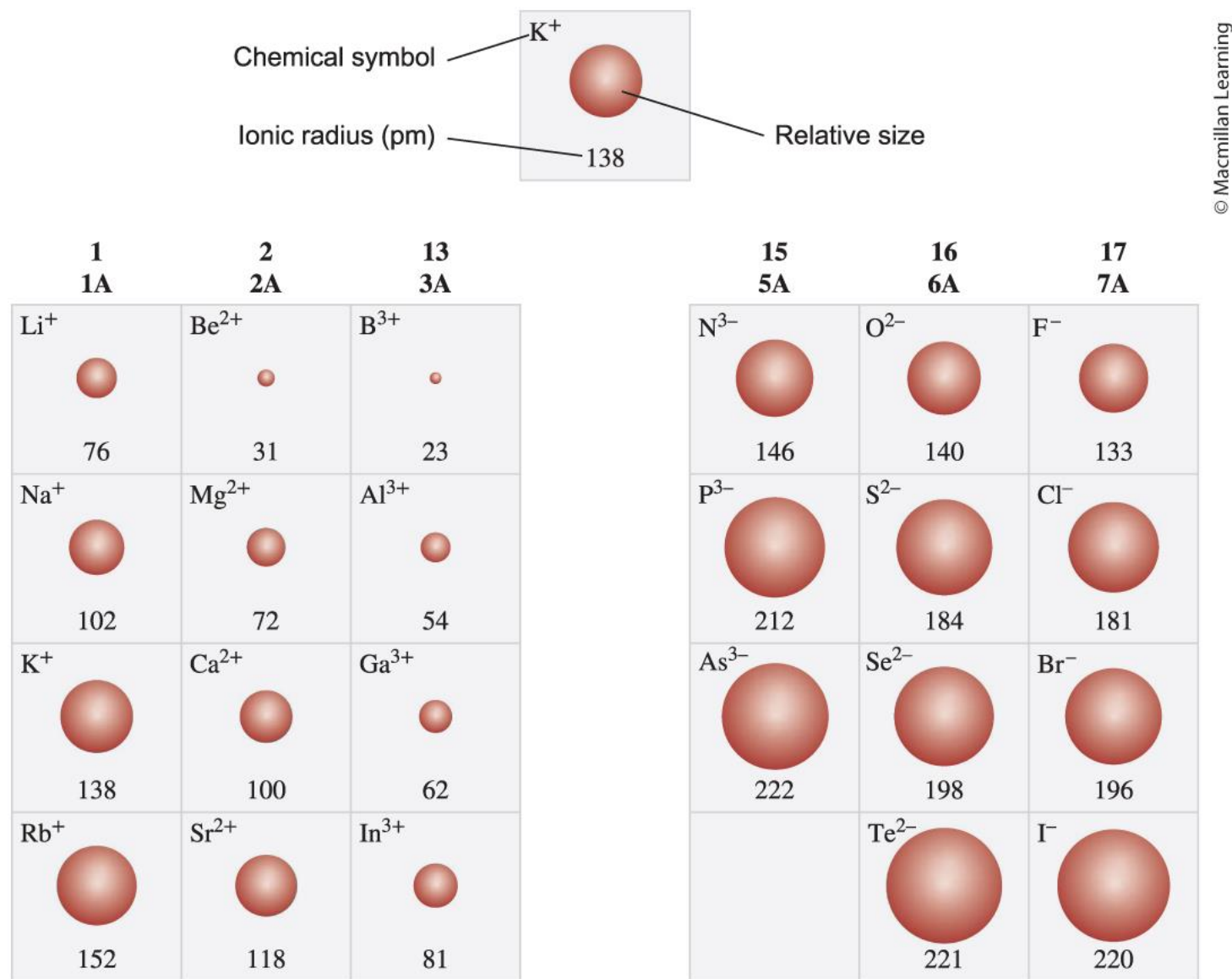


Figure 9.6. Periodic trends in ionic size in the main group.

Arrange these ions in order of increasing radius:

A. Ca^{2+}

B. Cl^-

C. S^{2-}

D. K^+

Enter your response as a string of letters from smallest to largest, such as “ABCD.”

Chapter 9.B

Periodicity and Ionic Bonding

Chemical Principles I

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9.3 Defining Ionization Energy



- Energies involved in adding an electron to or removing an electron from the atom follow periodic trends.
- **Ionization energy** is the energy input required to remove an electron from a gaseous atom to produce a gaseous cation and free electron.

9.3 Ionization Energies of the Elements

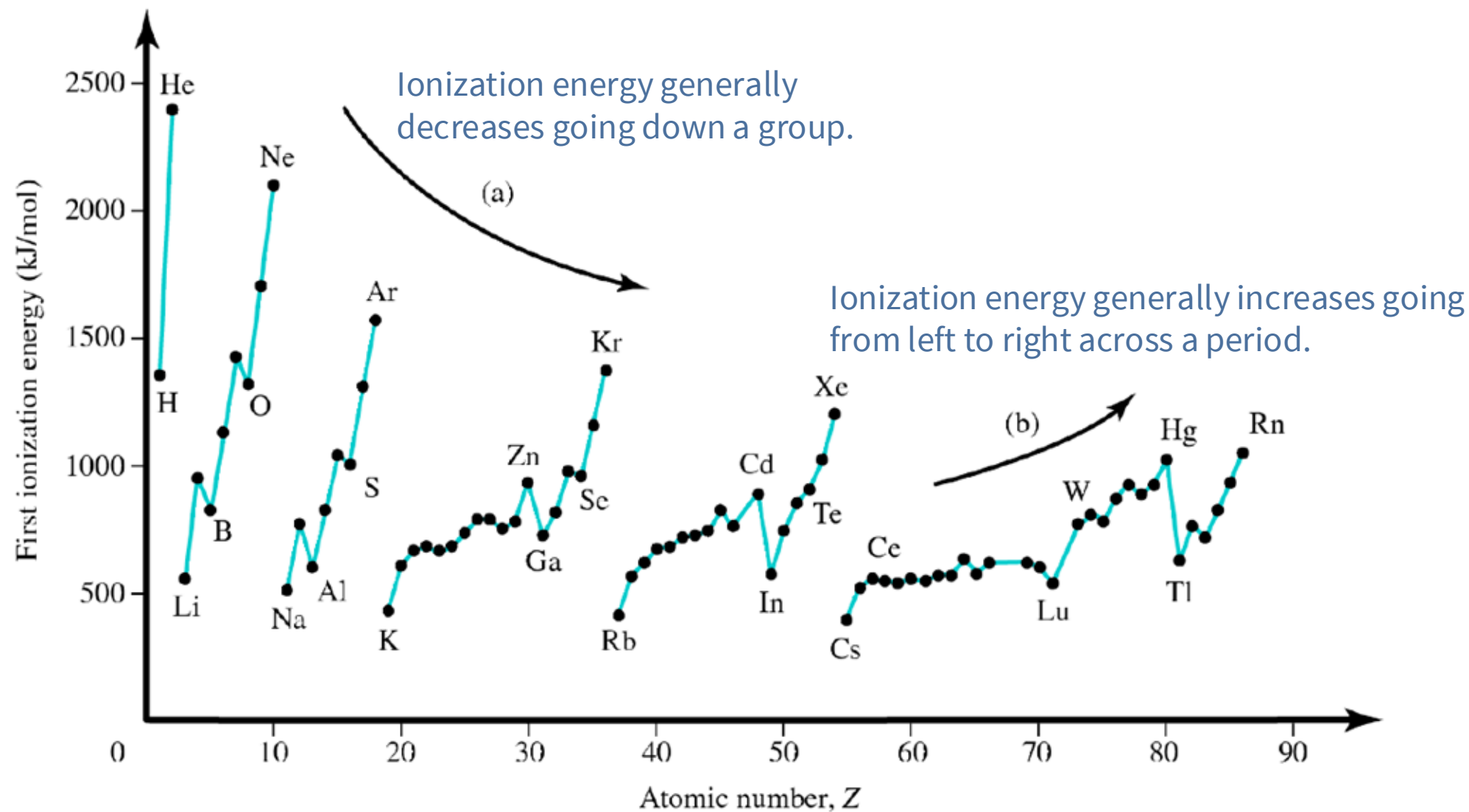


Figure 9.7. Ionization energies of the elements.

9.3 Periodic Trends in Ionization Energy

- Ionization energy and atomic radius are negatively correlated...

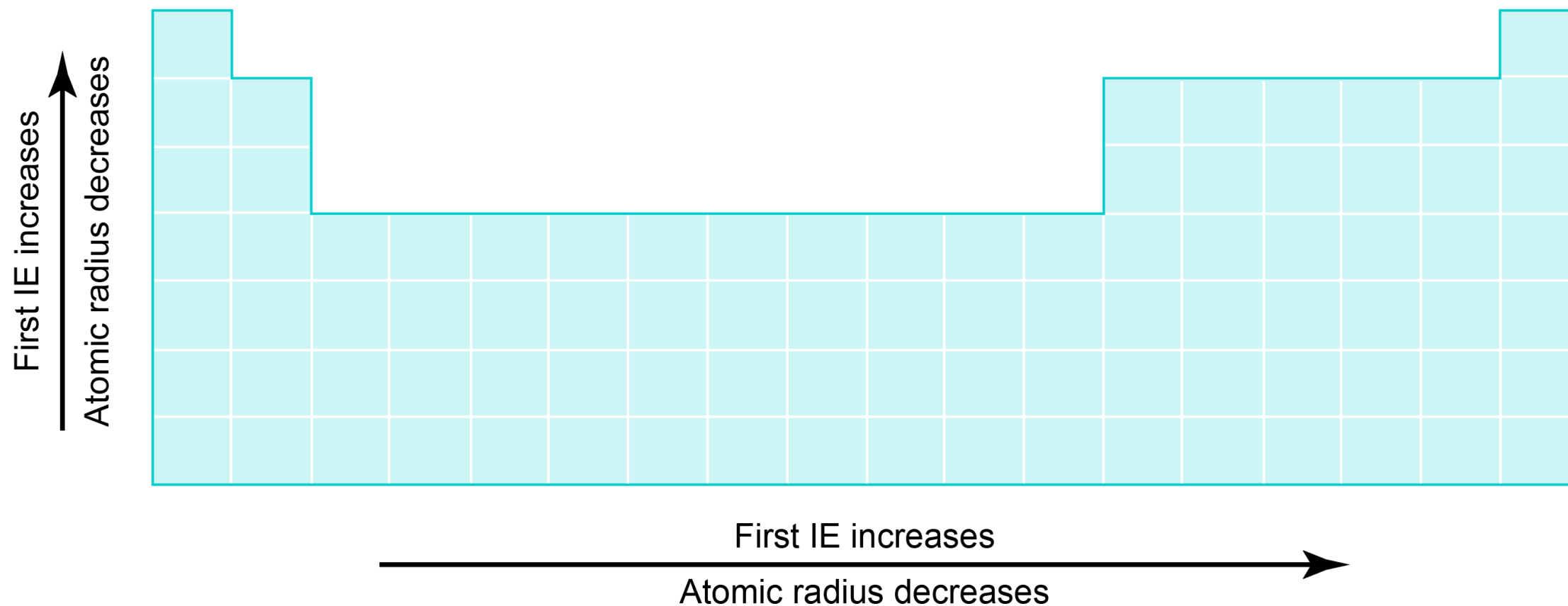


Figure 9.8. Trends in first ionization energy and atomic radius.

- Some exceptions to the general trend in ionization energy occur when the outermost valence electron is in a subshell by itself.
- **Example.** Beryllium (9.32 eV) versus boron (8.30 eV)

- Other exceptions involve elements containing only one set of paired electrons in the valence p subshell (group 16).
- **Example.** Nitrogen (14.5 eV) versus oxygen (13.6 eV)

Which statement is *true* regarding first ionization energy?
("First" refers to loss of the first electron from the neutral atom.)

- A. It represents the energy released when an electron is removed from a gas phase atom.
- B. The first ionization energy for lithium is greater than that for fluorine.
- C. First ionization energy corresponds to the process: $X(g) + e^- \rightarrow X^-(g) + IE$.
- D. The first ionization energy for lithium should be less than that for sodium because sodium has fewer core electrons.
- E. The first ionization energies for noble gases are very large because of their very stable electron configurations.

9.3 Higher Ionization Energies



- Higher ionization energies involve the removal of a second electron, third, fourth, etc.
- Because Z_{eff} increases significantly with each electron lost, $IE_1 < IE_2 < IE_3 < \dots$
- IE for the first core electron lost is ***much*** larger than the previous IE!

An unknown main group element in the fourth period experiences a large jump between the sixth and seventh ionization energies. What is the identity of the element? Enter its chemical symbol.

- **Electron affinity** is the change in energy when a neutral atom in the gas phase captures a free electron and forms a gaseous anion.



9.3 Periodic Trends in Electron Affinity

- Electron affinity predictably becomes less exothermic moving down a group, since the new electron is accepted into a shell farther from the nucleus.
- Electron affinity *generally* becomes more negative left to right across a period (ignoring the noble gases).

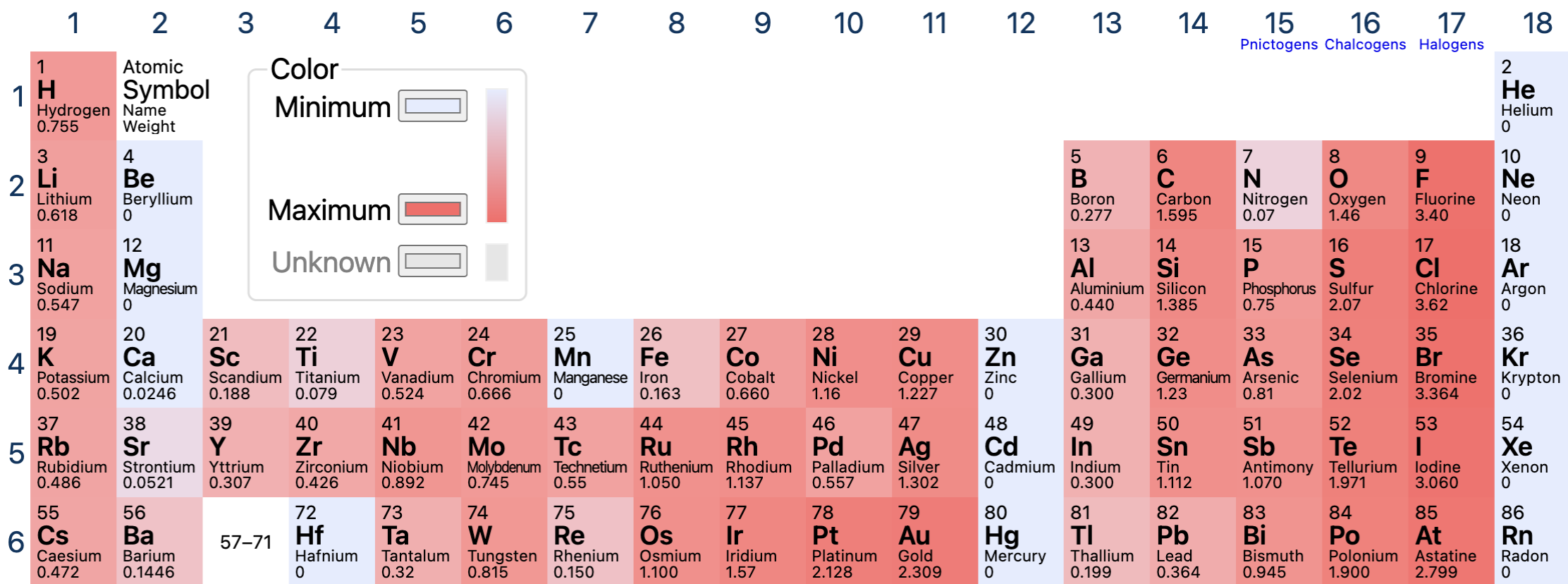
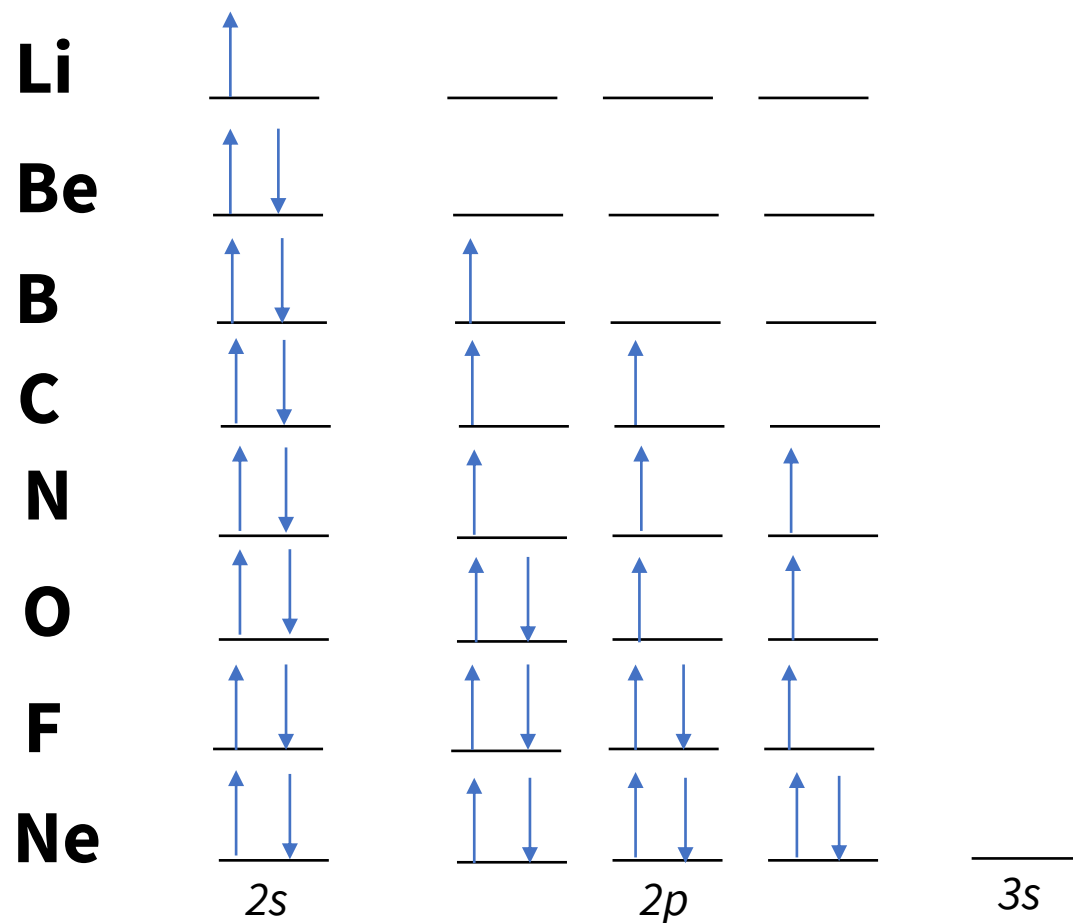


Figure 9.10. Electron affinities of the elements in electron-Volts (eV). A positive value indicates the quantity of energy released per electron captured; zero indicates endothermic electron capture.

9.3 Electron Configurations Explain Electron Affinity

- Why the numerous exceptions? Consider the electron configuration after accepting a new electron. For which elements are there “energetic penalties” associated with accepting another electron?

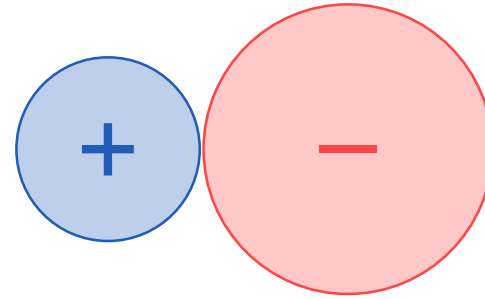


Rank the following elements from lowest (1) to highest (5) electron affinity. We will treat electron affinities as positive values—if it helps, read it as “affinity for electrons.” Greater affinity is associated with a higher number, as it should be.

- A. Li
- B. F
- C. N
- D. O
- E. Na

9.4 Ionic Bonding

- An **ionic bond** is the electrostatic attraction between oppositely charged ions.



- Ionic compounds contain an essentially infinite three-dimensional **lattice** of cations and anions.

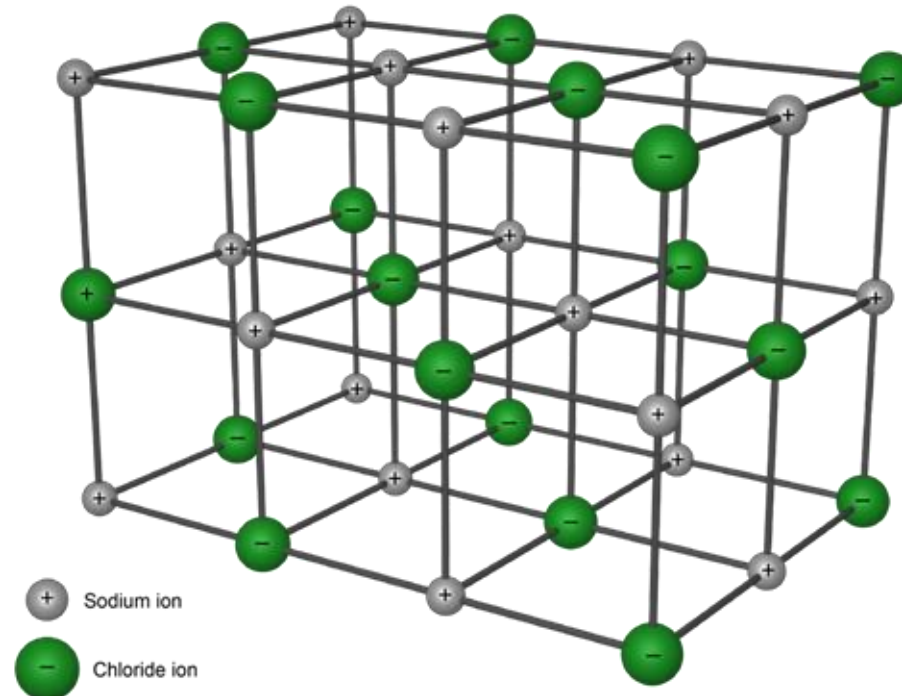
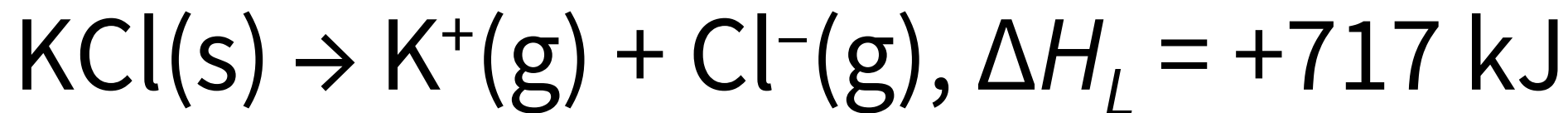


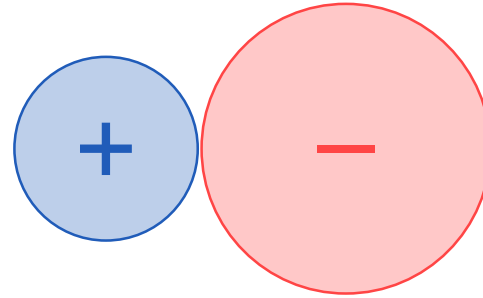
Figure 9.12. Ionic lattice in sodium chloride.

- Ionic bond strength can be measured as **lattice energy** (ΔH_L), the energy input required to “blast” 1 mole of an ionic lattice into gaseous ions.



9.5 Periodic Trends in Lattice Energy

- Trends in lattice energy can be accounted for by **Coulomb's law**.



9.5 Ionic Radius and Lattice Energy



- Smaller ions are associated with more stable lattices and higher lattice energy (shorter r).
- Charges greater in magnitude are associated with more stable lattices and higher lattice energy (greater q_1q_2)

Order the following compounds from largest (1) to smallest (5) lattice energy. Enter your response as a string of letters from largest to smallest lattice energy, such as “ABCDE.”

A. CsBr

B. SrO

C. CrN

D. MgO

E. Na₂S

9.5 Born-Haber Cycles

- Lattice energies are not directly measured, but instead are calculated by applying Hess's law to a series of connected processes called a **Born-Haber cycle**.

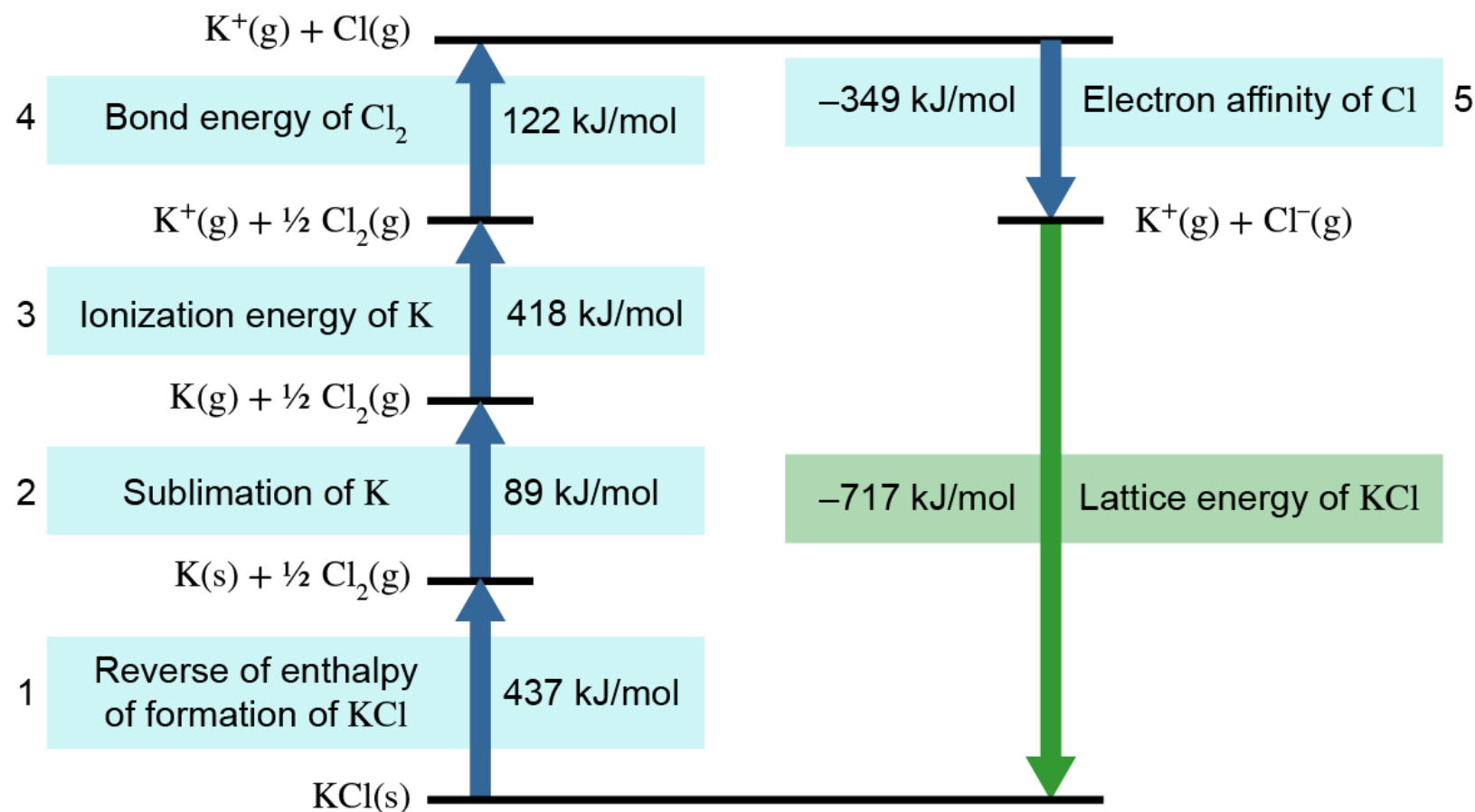


Figure 9.14. Born-Haber cycle for potassium chloride.

- Some of the amazing things we can now do with periodicity and ionic bonding...
 1. Determine the number of valence electrons in a neutral atom of a main group element.
 2. Calculate the effective nuclear charge felt by an electron in the valence subshell (Z_{eff}) given the identity of an atom or ion, using Slater's rules or related systems.
 3. Apply periodic trends in atomic and ionic radii to predict the relative sizes of atoms and/or ions; rationalize these trends using electron configuration and Z_{eff} .
 4. Apply periodic trends in ionization energy to predict the relative ionization energies of a series of elements.
 5. Apply periodic trends in electron affinity to predict the relative electron affinities of a series of elements.
 6. Rationalize trends in ionization energy and electron affinity (and exceptions) using electron configuration and Z_{eff} .
 7. Apply periodic trends in ionic radii and charges of monatomic ions to predict the relative lattice energies of a series of binary ionic compounds.